

Server Monitoring

Introduction

One of the most exciting aspects of working with Zimbra Collaboration Suite (ZCS) is the opportunity to engage with a diverse range of technologies, including Java, Postfix, OpenLDAP, and MySQL. To effectively manage and maintain a ZCS environment, administrators are encouraged to develop a strong foundational understanding of these components. This knowledge is essential not only for system monitoring but also for diagnosing and resolving performance-related issues.

ZCS actively gathers a wealth of performance metrics to assist administrators in maintaining optimal system health. These statistics are stored as CSV files within the `/opt/zimbra/zmstat` directory, providing valuable insights for analysis and troubleshooting.

The ZCS server continuously gathers a wide range of performance related statistics to support effective system management. These valuable data points are organized and stored as CSV files within the `/opt/zimbra/zmstat` directory, providing administrators with easy access for monitoring and analysis.

- **cpu.csv**: CPU utilization
- **fd.csv**: file descriptor count
- **mailboxd.csv**: ZCS server and JVM statistics
- **mtaqueue.csv**: Postfix queue
- **proc.csv**: disk utilization
- **soap.csv**: SOAP request processing time
- **threads.csv**: JVM thread counts
- **vm.csv**: Linux VM statistics (from the *vmstat* command)

These files are formatted as standard CSVs, making them easy to import into applications like Excel for detailed viewing and charting. Each day at midnight, the files are automatically archived into subdirectories within the `/opt/zimbra/zmstat` directory, ensuring organized and efficient historical data management.

What is Zmstats

Zmstats is Zimbra's powerful tool for bringing system performance and operational insights to life. Tracking everything from disk usage and CPU performance to Java

metrics and Zimbra counters, Zmstats delivers the critical insights you need to keep your system optimized, responsive, and operational.

Zmstats is made up of a powerful suite of components and scripts, each designed to work seamlessly together and provide a comprehensive view of your system's performance.

- `zmstatctl` (all `zmstat` collection scripts are located in `/opt/zimbra/libexec`)
 - `zmstat-allprocs` (since ZCS 6.0)
 - `zmstat-convertd`
 - `zmstat-cpu`
 - `zmstat-df` (since ZCS 6.0)
 - `zmstat-fd`
 - `zmstat-io`
 - `zmstat-ldap` (since ZCS 8.0)
 - `zmstat-mtaqueue`
 - `zmstat-mysql`
 - `zmstat-nginx` (since ZCS 6.0)
 - `zmstat-proc`
 - `zmstat-vm`
- `zmstat-chart`
 - `zmstat-chart-config`

Running zmstats

- Ensure that the `zmstat-chart.xml` file is available on the server where the stats were generated. When you run `zmdiaglog`, both the `zmstats` and `zmstat-chart-config` files are automatically created, streamlining the process for you.

```
# su - zimbra
```

```
$ zmstat-chart-config > /tmp/zmstat-chart-`zmhostname`.xml
```

- Make a charts directory:

```
$ mkdir ~/zmstat/2025-04-28/charts
```

- Then produce the stats:

```
$ zmstat-chart -c /tmp/zmstat-chart.xml -s ~/zmstat/2025-04-28 -d ~/zmstat/2025-04-28/charts
```

zmstatctl

zmstatctl is the command used to start and stop a variety of zmstat-* data logging scripts, giving you full control over the collection and management of system performance data.

zmstat-chart Utility

Zimbra offers a powerful command-line utility called `zmstat-chart` that transforms CSV data into clear, actionable visualizations. By leveraging an XML configuration file generated by `zmstat-chart-config`, it creates a series of HTML and PNG graphs. These dynamic charts allow administrators to quickly diagnose issues, assess system load, and gain valuable insights for proactive performance management.

```
$ zmstat-chart -s /opt/zimbra/zmstat/2025-04-28 -d ~/zmstat/2025-04-28/charts
```

The *zmstat-chart* utility reads data from CSV files located in `/opt/zimbra/2025-04-28/` and generates HTML and PNG charts, which are saved to the `~/zmstat/2025-04-28/charts/` directory (or `/opt/zimbra/zmstat/2025-04-28/charts/` if executed as the Zimbra user). The default chart settings can be found in the `/opt/zimbra/conf/zmstat-chart.xml` file.

If you wish to customize the charts, whether skipping specific ones or adding new ones not included by default, you can easily specify an alternate configuration file using the `-c` option.

Before running *zmstat-chart*, the configuration file may need to be generated.

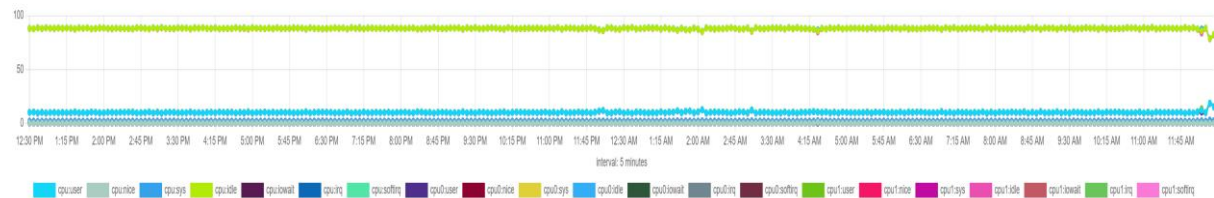
To do so, simply run:

```
$ zmstat-chart-config > /opt/zimbra/conf/zmstat-chart.xml
```

Chart Analysis

CPU utilization

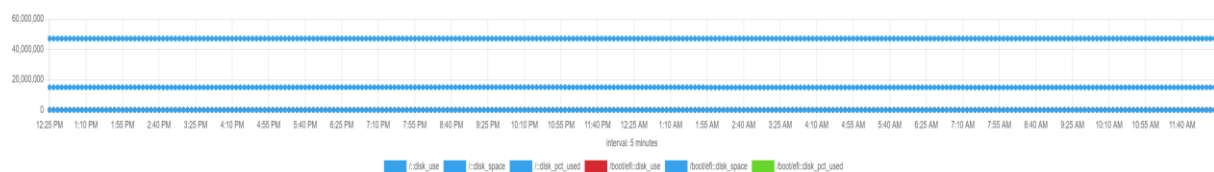
CPU utilization is monitored at both the server and process levels, providing a comprehensive view of system performance. Below is an example of a process CPU graph, offering valuable insights into how individual processes impact overall system load.



The chart shows a consistent server state with the CPU remaining idle. If a spike occurs, it will be recorded in the chart. To further investigate the cause, you can examine other charts or review the server logs for more detailed insights.

Disk Utilization

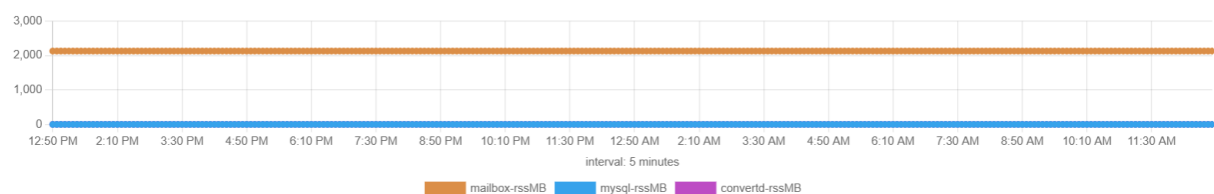
Disk utilization is monitored for each individual disk partition:



This chart highlights that disk activity rises in tandem with the increased CPU utilization.

Memory Consumption

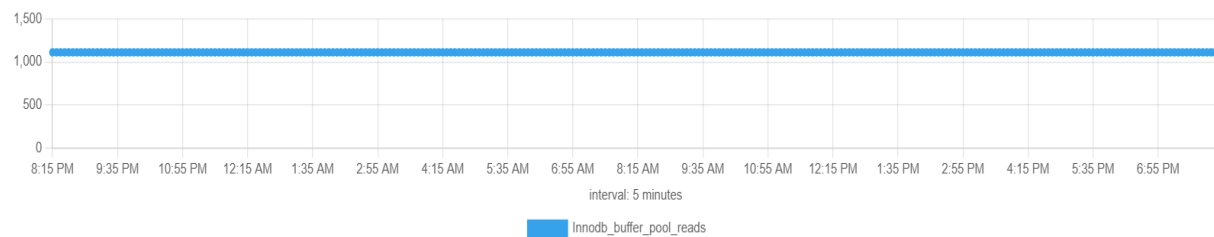
ZCS statistics monitor the memory usage of each process within the system:



This data helps in understanding how system memory is distributed across the different processes.

InnoDB Buffer Pool Hit Rate

This chart monitors the buffer pool hit rate for MySQL's InnoDB storage engine:



Higher buffer pool hit rates indicate that MySQL is successfully retrieving data from memory rather than from disk. If the hit rate falls below 990, it suggests that MySQL is relying too heavily on disk access. To address this, consider the following actions:

- Increase the buffer pool size in the *my.conf* configuration file.
- Use the EXPLAIN command on SQL statements found in */opt/zimbra/log/myslow.log* to identify queries that may be causing InnoDB to load large volumes of data into memory.

JVM Garbage Collection

ZCS monitors the percentage of time the Java Virtual Machine (JVM) dedicates to garbage collection.

zmstat-chart-config.xml

Controls what data is graphed and how it is presented.

Sample:

```
<chart title="Mailboxd: JVM Heap Used"
  category="mailboxd"
  infile="mailboxd.csv"
  outfile="mboxd-heap-used.png"
  yAxis="MB">
  <plot data="heap_used" legend="total" divisor="1m"/>
</chart>
```

The example above defines a chart that extracts the *"heap_used"* counter from mailboxd.csv and plots it into a file named *mboxd-heap-used.png*. The resulting graph will display *"heap_used"* over time, with the Y Axis labeled in megabytes (MB), where the values are calculated by dividing the counter by one million.

You can also add multiple plots to a single chart by including additional `<plot>` elements.

For more examples, you can refer to the `/opt/zimbra/conf/zmstat-chart.xml` file.